2.1. **Hanging rope**

A rope with length L and mass density per unit length ρ is suspended vertically from one end. Find the tension as a function of height along the rope.

2.2. **Block on a plane**

A block sits on a plane that is inclined at an angle θ. Assume that the friction force is large enough to keep the block at rest. What are the horizontal components of the friction and normal forces acting on the block? For what θ are these horizontal components maximum?